Jonas Wagner

Curriculum Vitae

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Education

Since Fall 2020 — **The University of Texas at Dallas**
PhD Mechanical Engineering (M.S. obtained in 2023)
Concentration: Dynamic Systems and Controls
GPA: 3.55

Fall 2016 - Spring 2020 — **University of Wisconsin-Platteville**
B.S. Engineering Physics and B.S. Electrical Engineering
Emphasis: Control Systems, Minor: Mathematics
GPA: 3.37

Teaching Experience

Fall 2026 — **Teaching Associate (Course Instructor) - Systems and Controls**
Mechanical Engineering, The University of Texas at Dallas
- Assigned to teach a large section (80 students) of the systems and controls lecture
- Collaborated with other instructor to develop custom course textbook, materials, and assessments
- Demos, exploratory learning tasks, and in-class working problems were developed to keep students engaged and combat the deskilling from improper generative AI usage
- Lessons continue to be learned throughout this experimental teaching process

Summer 2026 — **Teaching Associate (Course Instructor) - Systems and Controls Lab**
Mechanical Engineering, The University of Texas at Dallas
- Instructor for the first summer offering of the Systems and Controls Lab
- Held alternating weeks of lectures and labs for 2 course sections
- Modified course for a shorter summer semester
- Introduced in-lab quizzes to more accurately evaluate student learning in the age of AI

Spring 2026 — **Teaching Associate (Course Instructor) - Computer Aided Design Lab**
Mechanical Engineering, The University of Texas at Dallas
- Instructor for one section of the CAD Lab course on Creo Parametric (34 students)
- Lectured on foundational drafting principles and modeling techniques
- Guided students through tutorials and evaluated capabilities with in-class modeling assignments

Fall 2024/ Spring 2025 — **Teaching Associate (Course Instructor) - Systems and Controls Lab**
Mechanical Engineering, The University of Texas at Dallas
- Taught the senior-level systems and controls course lab with up to 128 students
- Held weekly lecture sessions preparing students for the labs
- Updated the course assignments and lab procedures to improve students learning outcomes
- Managed the 5 labs sections with the help of 2 TAs

Spring 2023 - Present — **Engineering Mentor - UTDesign EPICS**
Jonsson School of ECS, The University of Texas at Dallas
- Mentored multiple five-person teams of undergraduate students
- Assisted nonprofit sponsors in defining the project requirements
- Aided students on technical aspects of their projects

Fall 2023 **Teaching Assistant - Systems and Controls**
Mechanical Engineering, The University of Texas at Dallas
Professor: Dr. Yaoyu Li
- Managed weekly assignments and held bi-weekly recitation sessions
- Developed and managed an optional design-related final project
- Prepared and lectured the pre-exam in-class review sessions

Fall 2020 - Spring 2022 **Teaching Assistant - Introduction to Mechanical Engineering I & II**
Mechanical Engineering, The University of Texas at Dallas
Professors: Dr. Oziel Rios and Dr. Dani Fadda
- TA for 4 sections totaling more than 200 students
- Instructed students in person on weekly labs and assignments
- Graded weekly deliverable and answered any grading related questions
- Collected and aggregated assignments for ABET Accreditation
- Communicated with students via email and MS Teams to answer course-related questions

Fall 2019 - Spring 2020 **Lab Assistant - Intro to Engineering Projects/Intro to Automatic Controls**
Electrical and Computer Engineering, University of Wisconsin-Platteville
Professor: Dr. Mehdi Roopaei
- Supervised and insisted in instruction for lab courses
- Assisted in teaching first year undergraduate students through the Electrical Engineering Module
- Transitioned DC-motor control labs into virtual Simulink-based lab for use during COVID

Summer 2019 **Student Assistant - Online Course Development**
Center for Distance Learning, University of Wisconsin-Platteville
Professor: Dr. Mehdi Roopaei
Assisted in the development of course materials for the online graduate course:
Engineering 7310 - Control Systems Engineering I

Fall 2016 - Spring 2020 **Robot Design and Controls Mentor**
FIRST Robotics Competition Team 171, Platteville,WI
- Mentored High School students to design, built, and control robots for competition
- Teach fundamental math and physics concepts while inspiring students to pursue STEM careers
- Facilitate the logistics of traveling for competition and outreach events
- Restructured the club administration to allow expansion of the organization to additional STEM programs throughout the area K-12 education system

Relevant Coursework

Fall 2026 Engineering Design Team Coaching
Spring 2025 Deep Learning
Fall 2023 Model Predictive Control
Spring 2023 Optimal Control and Dynamic Programming
Fall 2022 Real Analysis
Spring 2022 Robust Control Systems · Multi-Agent Robotic Systems · Elementary Analysis II
Fall 2021 Engineering Optimization · Elementary Analysis I
Spring 2021 Nonlinear Systems · Convex Optimization · Dynamics of Complex Networks and Systems
Fall 2020 Linear Systems · Optimal Estimation & Kalman Filters · Probability & Random Variables
Spring 2020 Digital Signal Processing · Measurements and Instrumentation · Senior Design
Fall 2019 Discrete Time Controls · Electric and Magnetic Fields

Spring 2019	Modern Control Systems · Engineering Physics Sensors Lab · Analog Electronics
Fall 2018	Automatic Controls · Logic and Digital Design · Applied Mechanics
Spring 2018	Signals and Systems · Engineering Computation · Applied Optics

Awards

2022-Present	UTD Jonsson School - Dean's Teaching Fellowship
Fall 2025	UTD Homecoming Court - Royalty
Spring 2022	UTD Jonsson School - Best Teaching Assistant in Mechanical Engineering
Spring 2021	UTD Mechanical Engineering - Outstanding Contributions to Undergraduate Education
S/F 2019	Undergraduate Research, Scholastic and Creative Activity (URSCA) Scholarship
Spring 2019	UW Platteville Prototype Hackathon - 3rd Place
Spring 2019	Foxconn Smart Cities Smart Futures Competition - Winner (Round 1 & 2)

Affiliations

Since 2025	IEEE Control System Society, Student Member
Since 2020	IEEE, Student Member

Publications

J. Wagner and J. Ruths (2026a). "Encoding and Preserving Dependencies within Sets". In: *2026 IEEE 65th Conference on Decision and Control (CDC)*.

– (2026b). "Dependency Encoding within Set-Based State Estimation". In: *2026 American Control Conference (ACC)*.

– (Under Review). "Set-based Anomaly Detector with Safety Guarantees". In: *IEEE Transactions on Automatic Control*.

T. Kogal, **J. Wagner**, and J. Ruths (2025). "safeSLAM: Trustworthy Localization via Zonotopic Memory". In: *2025 IEEE 64th Conference on Decision and Control (CDC)*. IEEE, pp. 7076–7082.

J. Wagner and J. Ruths (2025). "memZono: An Extension to zonoLAB for Dependency-Preserving and Dimensionally-Aware Set Operations". In: *2025 American Control Conference (ACC)*.

J. Wagner, T. Kogel, and J. Ruths (2023). "Set-Based Anomaly Detector and Stealthy Attack Impact Using Constrained Zonotopes". In: *2023 62nd IEEE Conference on Decision and Control (CDC)*.

J. Wagner and M. Roopaei (2020). "Edge Based Decision Making in Disaster Response Systems". In: *IEEE - 10th Annual Computing and Communications Workshop and Conference*.

A. Fowler, E. Mutschelknaus, M. Roopaei, and **J. Wagner** (2019). "Learning in The Virtual Realm: A Platform for Immersive Engineering Education". In: *International Journal of Advances in Electronics and Computer Science (IJAECS)*.

Undergraduate Research Experience

Fall 2018 - Spring 2020

Projects Involving Machine Learning and Virtual Reality
Electrical and Computer Engineering, University of Wisconsin-Platteville
Advisor: Dr. Mehdi Roopaei

Disaster Response Applications (ML, Edge Analytics, and VR)
- Wrote several grant proposals (approx. $15 K awarded) that funded research into the use of ML and edge analytics within a multi-agent framework for disaster response
- Developed a virtual framework to develop and test an object detection algorithm
- Working on training a neural network using the Darknet framework to perform object detection on a custom database
- Submitted a manuscript detailing this virtual framework to the IEEE 10th Annual Computing and Communication Workshop and Conference

Computer Vision at the Edge on a Jetson Nano
- Explored the Jetson Nano Platform and worked within a Linux environment
- Used existing tools to connect a CSI camera and detect faces using Haar classifiers

Applying VR to Education
- Assisted in the preliminary development of a VR framework for distance education
- Assisted other students in creating a dynamic system visualization platform to provide students with an interactive environment to understand dynamic system modeling

Exploring Unity ML Agents
- Worked with Unity ML Agents to learn about ML and reinforcement learning methods
- Used pre-trained ML models and explored how well agents could perform the same objective in modified virtual environments

Spring 2019 - Spring 2020

Implementing K-Means and EM-Algorithm in MATLAB and Python
Electrical and Computer Engineering, University of Wisconsin-Platteville
Advisor: Dr. Hynek Boril

Learned about fundamental statistical modeling and ML techniques while also learning Python
- Implemented K-means Clustering and the EM-Algorithm to statistically model data
- Used Windows Subsystem for Linux and Midnight Commander to run Python naively

Fall 2018 - Summer 2020

Computational Analysis of MEMS Pressure Sensors
Engineering Physics, University of Wisconsin-Platteville
Advisor: Dr. Gokul Gopalakrishnan

Evaluated the limitations of different methods used for modeling the behavior of silicon nanomembranes for MEMS pressure sensing applications
- Focused primarily on automating the computation and analysis process
- Used ANSYS Workbench to perform FEM analysis on single crystalline silicon membranes under uniform pressure
- Used Python (NumPy and pandas) to automate data analysis
- Created plots to visualize data with matplotlib

June 2019

LabVIEW Programming of a Mobile Robot
Mechanical Engineering, University of the West of Scotland - Paisley
Advisor: Dr. Luc Rolland

Short-term study abroad research trip: Worked on developing a control algorithm for a sbRIO controlled robot that avoids obstacles and maps an environment autonomously

Industry Experience

May 2018 - Aug 2018 **Summer Employee - Blown Film Department**
Bemis Converter Films, Oshkosh, WI
- Helped operators of Blown Film Machines during 12-hour shifts
- Operated forklift to transport supplies and waste
- Filed reports to insure quality and accuracy of film composition

May 2017 - Aug 2017 **Summer Employee - Press Department**
Bemis Specialty Films, Oshkosh, WI
- Assisted in mounting for flexographic printing presses
- Operated Tug to transport flexographic press cylinders
- Organized mounting priorities for 12 machines during 12-hour shifts

Community Engagement

Spring 2022 - Present **Committee Member**
Center for Students and Scolars in Controls (CSSC)
- Assist in developing and hosting the grad-student led control systems workshops each semester
- Participated in the weekly (primarily student-led) DSC seminars
- As Secretary (Fall 2025-Spring 2026) I led weekly meetings and compiled the monthly newsletter
- Prepared and taught Latex (overleaf) worship in Fall 2022

Spring 2024 - Present **Committee Member**
Mechanical Engineering Graduate Student Association (MEGA) at UTD

Spring 2022 - Fall 2024 **Graduate Student Mentor**
Jonnsson School of Engineering and Computer Science

Fall 2022 - Spring 2023 **Committee Member**
Strategic Planning Committee, Jonsson School of Engineering and Computer Science

Fall 2020 - Present **Taskmaster** (Secretary/VP/Committee Chair)
For Autistic Empowerment (FAE) @ UTD
- Ensure all important administrative tasks are attended to during weekly meetings
- Led the Constitution Committee during club restructuring (F24) and bylaw updates (F26)
- Organized travel for in-person conferences (S22/S23)
- Represent FAE (and Wesley@UTD in 2023-2026) on campus through the Rainbow Coalition
- Guide other students through life's struggles as effectively the club elder

Since 2026 Arapaho UMC Choir

Fall 2022 - Spring 2026 **Worship Band Leader**
Synergy Wesley Foundation/Wesley@UTD, Richardson, TX
- Helped lead worship (vocal/keyboard) during weekly "Wesley Wednesday"
- Organized and supervised "Synergy Social" events
- Represent Wesley throughout DFW through Black Tie Dinner Events

Since 2021 Credo Community Choir